

Dexter Fernandes

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Machine Learning Engineer

Ambitious SLAM Engineer with hands-on expertise in developing and deploying LiDAR-based RTAB-Map pipelines and waypoint navigation in Gazebo-simulated and real-world environments. Proven ability to collaborate effectively in cross-functional teams, communicate complex technical concepts clearly, and thrive in fast-paced startup environments. At the University of Bristol, delivered robust localisation and mapping performance by optimising SLAM parameters for both indoor high-detail and outdoor feature-scarce settings, and successfully integrated gamma-ray sensor fusion for real-time point-cloud annotation.

WORK EXPERIENCE

Esbaa • Muscat, Oman

01/2023 - 08/2023

AI Engineer

- Developed time-series machine learning models for the classification of on-shore oil rig operations to increase worker safety and reduce supervision required during 24/7 operation, achieving a 23% higher F1 score than human experts.
- Led strategic MLOps initiatives to continuously enhance experimentation and deployment workflows through bayesian hyperparameter tuning of time-series and vision models.
- Spearheaded the end-to-end development and production rollout of YOLOv5 models, to elevate rig worker and equipment detection precision by 15% in 3 months.
- Refactored a legacy monolithic system into a Kafka-driven microservice architecture, containerising each service with Docker to enhance failure isolation and improve MTBF by 35% within four months
- Designed a dynamic memory management solution to address recurring crashes in a live video streaming system, reducing crash frequency by 67%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

05/2020 - 12/2022

Lead Software Engineer - AI

- Led a 4-engineer team on ViCAS to design, train, and deploy deep learning models for real-time object detection, pose estimation, and tracking on edge hardware to reduce false positives by 33% while achieving sub-50 ms inference latency within 6 months.
- Developed core perception algorithms on resource-constrained Intel edge devices, leveraging mixed-precision and fixed-point quantization with structured pruning to meet strict memory and latency budgets for real-time decision-making in industrial environments.
- Implemented rigorous testing and validation frameworks based on simulated adverse conditions, to ensure model accuracy and stability, reducing performance variance by 25% under challenging lighting, occlusion, and noise scenarios.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

09/2023 - 09/2024

- Developed and implemented a novel real-time SLAM solution by integrating LiDAR-based mapping (RTAB-Map) with gamma-ray detection on an Unmanned Ground Vehicle (UGV).
- Conducted extensive tests to evaluate mapping accuracy using Velodyne VLP-16 LiDAR, achieving reliable performance in diverse environments, including indoor high-detail and outdoor feature-scarce areas.
- Optimised RTAB-Map parameters specifically for enhanced performance in varying terrain conditions, significantly improving the accuracy of localisation and mapping in challenging outdoor settings.
- Conducted simulations for 2D and 3D SLAM, as well as waypoint navigation of the Clearpath Husky robot in Gazebo, using ROS2 Humble to validate and tune navigation algorithms in both indoor and outdoor scenarios
- Validated the system through rigorous real-world experimentation, including the deployment of the robot in complex industrial environments, effectively navigating obstacles and identifying radioactive sources in real-time.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

07/2016 - 05/2020

SKILLS

- **Programming Languages:** C++, Python
- **Perception & Robotics:** GStreamer, LiDAR SLAM, Nav2, OpenCV, OpenVINO, ROS1 (Noetic), ROS2 (Humble/Foxy), Vision Transformer (ViT), YOLOv5/v8
- **Backend & Cloud / Edge:** Apache Kafka, AWS, FastAPI, Intel VPU, Microsoft Azure, NVIDIA Jetson AGX, PostgreSQL
- **DevOps & MLOps:** Docker, DVC (Data Version Control), Git, Linux, MLFlow, Optuna
- **AI Frameworks:** PyTorch, Sci-kit Learn, TensorFlow